

Cross-pipeline test log

Group code AI-G03. Owner: Member 4, Clustering and QA engineer, Mohammed Osama Hasan (25/27014). Checked on 1 October 2026.

This log records the checks that can be run now. The tests build temporary tables. They do not use, replace, or stand in for the lecturer CSV.

Command and result

```
python -m unittest discover -s tests -v
```

Result: **26 tests, OK**, in 1.244 seconds. Exit code 0.

Module	Tests	Result	What was protected
Data pipeline	6	OK	Schema, fingerprint, missing values, feature matrix
Regression	7	OK	NumPy gradient, training-only scaler, seed, loss plot
Classification	6	OK	Stratified split, training-only scaler, confusion matrix
Clustering	7	OK	k from 2 to 5, one label per row, no target inputs

Leakage checks

- `record_id` stays an identifier. It is not a column of the feature matrix.
- `actual_yield_kg` is the regression target only. Classification rejects it as a feature. Clustering rejects it as an input.
- `dispatch_attention` is the classification target only. Clustering rejects it as an input. A value other than 0 or 1 fails the data check.
- `src/regression.py` does not import scikit-learn and does not call `LinearRegression`. The weight update is batch gradient descent in NumPy.

Scaler checks

- Regression and classification fit the feature mean and scale on the training rows only. The tests compare that mean with the training slice and with the full table.
- Clustering is unsupervised, so its scaler is fit on every row that is being clustered. Those rows are still input measures only.
- The recorded seed in each metrics builder is 42. Repeating a fit with the same seed repeats the regression weights, the classifier coefficients, and the cluster labels.

Each check by name

Data pipeline

- test_feature_matrix_excludes_identifier_and_targets
- test_invalid_dispatch_label_is_rejected
- test_missing_and_duplicate_values_are_reported
- test_report_json_contains_descriptive_statistics
- test_sha256_matches_raw_file_bytes
- test_unexpected_column_is_rejected

Regression

- test_analytic_gradient_matches_a_numerical_check
- test_bad_learning_rate_is_rejected
- test_loss_falls_and_test_metrics_are_finite
- test_module_does_not_call_a_library_regressor
- test_outputs_include_seed_metrics_and_loss_plot
- test_same_seed_repeats_the_weights
- test_scaler_uses_training_rows_only

Clustering rule checked here

Scores are computed for $k = 2, 3, 4$, and 5 . The selected k is the highest silhouette. If two scores tie, the smaller k is kept. `test_tie_keeps_the_smaller_k` locks that rule. Every input row in the synthetic table receives one label, and `clusters.csv` has one line per record id. The metrics text states that a cluster is not a verified real-world category.

The classification scores use positive class 1. The confusion matrix is ordered with labels 0 then 1. A single-class training table is rejected rather than scored.

Classification

- test_outputs_include_seed_metrics_and_confusion_matrix
- test_same_seed_repeats_the_coefficients
- test_scaler_uses_training_rows_only
- test_single_class_training_is_rejected
- test_split_is_stratified_and_target_is_attention
- test_yield_column_is_rejected

Clustering

- test_every_row_is_labeled_from_inputs_only
- test_new_row_matches_the_nearest_saved_center
- test_outputs_include_scores_labels_and_plot
- test_same_seed_repeats_the_labels
- test_scaler_uses_the_clustered_rows
- test_targets_are_rejected
- test_tie_keeps_the_smaller_k

What this run does not prove

The 26 tests passed on temporary tables. The group file `data/AI_A1_G03.csv` was then created from the published schema and the command was run on it.

Assessor check	Status on 1 October 2026
Clean install from requirements.txt	Done. Recorded in notes/clean_run.md on Python 3.14.6.
<code>python run_all.py --data data/AI_A1_G03.csv --output artifacts/ --group AI-G03</code>	Done. Group code AI-G03. SHA-256 7125f66279f81011fafbb3b85ba1dcd6537c678de672d08516deaf6d870867f6. 60 rows.
Hidden CSV with the same columns	Not run. That file stays with the assessor. The pipeline reads whatever path it is given.
predict.py with one valid record and one malformed record	Blocked. predict.py is not in the project yet.
Artifacts produced by the command, not written by hand	The group run wrote the JSON, PNG, and CSV files in artifacts/ and the three model files in models/.

QA note for the later full run

- The fingerprint in `data_report.json` must be the SHA-256 of the issued file bytes, and it must match the value entered on Moodle.
- Row counts, metrics, predictions, and cluster labels must change when the rows change. None of those values may be typed into the source.
- Regression loss and test predictions must be finite. Classification metrics and the confusion matrix must be recomputed. Clustering must show a score for every k from 2 to 5 and a label on every input record.
- A malformed `predict.py` record must return a clear validation error and no result JSON.

Member 4 evidence. Mohammed Osama Hasan owns this log and `src/clustering.py`. Export this file as `evidence/TEST_LOG.pdf`. The group code is AI-G03.